REMARK ON OPENAI'S $\omega \le 9/4$: THE PROFILE AXIOMS STOP EXACTLY AT $9/4$, AND AN ENLARGED CONSTRAINT SYSTEM STILL DOES

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Abstract. OpenAI's preprint "An Upper Bound of 9/4 for the Matrix Multiplication Exponent" (openai/math, family 107) obtains $\omega \le 9/4$ from six properties (positivity, symmetry, boundary values, concavity, tripling, and a rank bound) of a symmetrized profile $P(a,b)$ of the polynomial-multiplication tensors. We give a self-contained proof that these six properties cannot yield any exponent below 9/4. The profile $P^*(a,b) = \phi(m)\big(M + \frac{m-1}{2}\big)$, with $m = \min(a,b)$, $M = \max(a,b)$ and $\phi(m) = \prod_{j<m}(1 + \frac{1}{3j})$, is the pointwise least profile satisfying the first five properties. It satisfies the rank bound exactly at $t = 3/4$. The least-element observation itself is not new: it was recorded publicly on 2026-10-07 in the GitHub repository `eliottcassidy2000/math` (THM-4568). It was also noted numerically on 2026-10-08 in `GasStationManager/SafeVerifyAgent`. Our proof of the rank bound uses only the elementary inequality $\phi(m)^3 \le 2m - 1$.

The second part reports joint work with an OpenAI Codex session. A *balanced assignment* $\lambda^*(C(a,b)^\pi) = P^*(a,b)^{3/4}$, $\lambda^*(\mathrm{dot}_m) = m^{3/4}$, $\lambda^*(T_m) = m^{9/4}$ satisfies, at all sizes, a substantially larger system of valid character inequalities. These include several new degenerations of polynomial multiplication. Hence that system cannot force $\omega < 9/4$ either. We state precisely which families are proved for all sizes, which are checked only on finite ranges, and which routes remain open. λ^* is *not* claimed to be a tensor character.

1. Setting and attribution

All tensor constructions and the proof of $\omega \le 9/4$ are OpenAI's work [1]. We quote line numbers of `build/paper.tex` at openai/math commit `adc7f1241b42`, which is unchanged at `08fcd618213a`. For a character λ with mean dot-product exponent $t = (p_X + p_Y + p_Z)/3$, the paper defines the symmetrized profile of $C(a,b) = \sum_{i<a,j<b} x_i y_j z_{i+j}$ (l. 420–422):

$$P(a,b) = \Big(\prod_{\pi \in S_3} \lambda_\pi(C(a,b)) \Big)^{1/(6t)}.$$

The passage from characters to the exponent uses exactly the following properties of P and nothing else.

- (Pos) $P(a,b) > 0$ (l. 427, l. 611).
- (S) $P(a,b) = P(b,a)$ (l. 427, l. 611).
- (B) $P(1,b) = b$ (l. 427, l. 611).
- (C) $2P(a,b) \ge P(a,b+1) + P(a,b-1)$ for $a \ge 1, b \ge 2$ (Lemma 4.1, l. 442–447; hypothesis at l. 612).
- (T) $P(a, 3h + a - 1) \ge 3P(a,h)$ for $a, h \ge 1$ (Lemma 4.2, l. 526–530; hypothesis at l. 613).
- (U_t) $P(a,b) \le (a+b-1)^{1/t}$ (l. 428). Only the diagonal case $P(a,a) \le (2a-1)^{1/t}$ is used (l. 657).

Lemma 5.1 (Diagonal growth, l. 610–650) takes (Pos), (S), (B), (C), (T) as its only hypotheses. The proof of the main theorem (l. 652–660) adds only (U_t). The character-existence lemma (l. 198–202, Appendix A) is used afterwards, only to pass from "$t \le 3/4$ for every character" to $\nu \le 9/4$. It places no further constraint on P.

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The Lean formalization in the same repository has the same structure. `structureScalarProfile` has exactly the six fields above (`Growth/Profile.lean`, l. 23–33). The theorem `meanExponent_le_three_quarters` reaches the exponent only through `toScalarProfile` and `ScalarProfile.exponent_le_three_quarters` (`Polynomial/Inequalities.lean`, l. 97–114). These paths are relative to `lean/OAI/LinearAlgebra/MatrixMultiplication/AuxiliarySeparation/`. *Related work.* Independent formal checks of [1] include [4], which extends the proof to all fields, and [3]. The latter also observes numerically (§2d) that the least feasible diagonal equals the bound in the paper's proof. Earlier "limits of the method" results concern the laser method and Coppersmith–Winograd tensors [9, 10, 8]. They do not apply to the profile argument, which is built on the spectral framework [6, 7, 11]. This remark complements [4]: [4] shows that the $9/4$ argument works over every field, and we show that its profile axioms cannot go below $9/4$.

We write $\mathrm{Ax}(t)$ for the system (Pos), (S), (B), (C), (T), (U_t). Every character satisfies $1/3 \le t \le 1$ (l. 211–234).

2. The least profile

Put $\phi(1) = 1$, $\phi(m) = \prod_{j=1}^{m-1}\big(1+\frac{1}{3j}\big) = \frac{\Gamma(m+1/3)}{\Gamma(4/3)\Gamma(m)}$, and

$$P^*(a,b) = \phi(m)\Big(M + \frac{m-1}{2}\Big), \qquad m = \min(a,b),\ M = \max(a,b).$$

Its diagonal values are $1, \frac{10}{3}, \frac{56}{9}, \frac{770}{81}, \dots$, and $P^*(a,a) = \frac{3a-1}{2}\phi(a) \sim \frac{3}{2\Gamma(4/3)}a^{4/3}$.

Theorem 1 (least profile; recorded earlier in [2])**.** *The following hold.*

(i) *P^* satisfies (Pos), (S), (B), (C), (T), and also ($U_{3/4}$). Equality holds in ($U_{3/4}$) only at $(1,1)$.*
(ii) *Every P satisfying (Pos), (S), (B), (C), (T) satisfies $P \ge P^*$ pointwise.*
(iii) *Hence, for $t > 0$, $\mathrm{Ax}(t)$ is satisfiable if and only if $t \le 3/4$. In particular no argument that uses only these properties of the profile can give $t < 3/4$, that is, $\omega < 9/4$. Every inequality in the proof of Lemma 5.1, up to and including (1), is an equality for P^*.*

Proof. (i) (Pos), (S) and (B) are immediate.

(C). Fix a and put $f(b) = P^*(a,b)$. For $b \ge a$, f is linear with slope $\phi(a)$. For $b+1 \le a$, the increment is $\delta(b) = f(b+1) - f(b) = \phi(b)\big(\frac{a}{3b} + \frac{2}{3}\big)$. The inequality $\delta(b+1) \le \delta(b)$ is equivalent to $(3b+1)(a+2b+2) \le (a+2b)(3b+3)$, which is $2(a-b-1) \ge 0$. This holds for $b \le a-2$, which is the range needed. At the junction, $\delta(a-1) = \phi(a-1)\frac{3a-2}{3(a-1)} = \phi(a)$, so the increments are non-increasing everywhere. They are also positive.

(T). Since $3h + a - 1 \ge a$, the left side is $\phi(a)\big(3h + \frac{3}{2}(a-1)\big)$. If $h \ge a$, the right side is $3\phi(a)(h + \frac{a-1}{2})$, so equality holds. If $h < a$, put $L(a') = \phi(a')(3h + \frac{3}{2}(a'-1))$ and $R(a') = 3\phi(h)(a' + \frac{h-1}{2})$ for $a' \ge h$. Then $L(h) = R(h)$, and

$$L(a'+1) - L(a') = \phi(a')\Big(2 + \frac{h}{a'}\Big), \qquad R(a'+1) - R(a') = 3\phi(h).$$

At $a' = h$ the two increments are equal. The left increment is non-decreasing in a': its forward difference is $\phi(a')\frac{2(a'+1-h)}{3a'(a'+1)} \ge 0$. Hence $L(a) \ge R(a)$, with equality exactly when $h \ge a-1$.

($U_{3/4}$). By induction, $\phi(m)^3 \le 2m-1$. The base case is $m = 1$. The inductive step $(1 + \frac{1}{3m})^3 \le \frac{2m+1}{2m-1}$ is equivalent to $(2m+1)(3m)^3 - (2m-1)(3m+1)^3 = 9m^2 + 7m + 1 > 0$. Since $M + m - 1 \ge M + \frac{m-1}{2}$ and $M + m - 1 \ge 2m - 1$,

$$P^*(a,b) \le \Big(M + \frac{m-1}{2}\Big)(2m-1)^{1/3} \le (a+b-1)^{4/3}.$$

For $m \ge 2$ the first inequality is strict. For $m = 1$ and $M \ge 2$ the second is strict. Alternatively, Wendel's inequality $\Gamma(m+\frac{1}{3})/\Gamma(m) \le m^{1/3}$ [12] gives $\phi(m)^3 \le \Gamma(4/3)^{-3} m \approx 1.4043\, m \le 2m-1$ for $m \ge 2$.

(ii) We follow the paper's own chain (l. 617–639). For fixed a, (C) makes $\Delta_{a,h} = P(a,h+1) - P(a,h)$ non-increasing in h. Write $D_a = P(a,a)$. Lemma 5.1's proof gives

$$D_a \;\ge\; \frac{3a-1}{2} \prod_{m=1}^{a-1} \Big(1 + \frac{1}{3m}\Big) = P^*(a,a). \tag{1}$$

Off the diagonal, take $b > a$ and $h \ge a$. By (T) and (C),

$$2P(a,h) \le P(a,3h+a-1) - P(a,h) = \sum_{k=h}^{3h+a-2} \Delta_{a,k} \le (2h+a-1)\Delta_{a,h},$$

so $P(a,h+1) \ge P(a,h)\frac{2h+a+1}{2h+a-1}$. Telescoping from $h = a$ to $b-1$ gives

$$P(a,b) \ge D_a \cdot \frac{2b+a-1}{3a-1} \ge P^*(a,b).$$

By (S) the same bound holds for $b < a$. This off-diagonal step was also given by an auditor in [2] and in [5, Codex round 3, (1)].

(iii) If $0 < t \le 3/4$, then $(a+b-1)^{1/t} \ge (a+b-1)^{4/3}$, so P^* satisfies Ax(t) by (i). Conversely, Lemma 5.1 gives $a^{4/3} \le P(a,a) \le (2a-1)^{1/t}$ for every a, which forces $t \le 3/4$. The tightness statements follow from the equality cases in (i): P^* satisfies $\Delta_{a,a} = \phi(a)$, equality in (T) at $h = a$, and equality in the diagonal recursion. Only the last two simplifications in the paper, $(1+\frac{1}{3m})^3 \ge 1 + \frac{1}{m}$ and $\frac{3a-1}{2} \ge a$, are strict. They lose the constant $\frac{3}{2\Gamma(4/3)} \approx 1.68$ but not the exponent. □

3. What the barrier does and does not say

(1) It concerns the *axiom system* Ax(t), not tensors. Theorem 1 does not say that some character has profile P^*. It does not say that $\omega = 9/4$. It does not say that polynomial multiplication tensors can never prove $\omega < 9/4$.

(2) To go below 9/4 with this approach, one needs a further valid inequality for characters that every profile in the feasible set at $t = 3/4$ violates. That set contains P^* and, by (ii), only profiles $\ge P^*$. Alternatively, one needs information that the symmetrized profile does not see.

(3) The formal values below are not claimed to come from a character. If some character λ had $\lambda(T_m) = m^{9/4}$ for some $m \ge 2$, then by multiplicativity and $\lambda \le \mathrm{R}$ we would get $\mathrm{R}(T_{m^k}) \ge m^{9k/4}$ for all k, so $\omega \ge 9/4$. Together with [1] this would give $\omega = 9/4$; compare Strassen's spectral characterisation of ω [6]. Nothing here proves or refutes this.

4. An enlarged constraint system (joint work with OpenAI Codex)

In three rounds on 2026-10-09, an OpenAI Codex session (model `gpt-6-astra`, Codex CLI v0.157.0) and we searched for *valid* character inequalities that might exclude the balanced assignment

$$p_X = p_Y = p_Z = t = \tfrac{3}{4}, \qquad \lambda^*(C(a,b)^\pi) = P^*(a,b)^{3/4}\ (\pi \in S_3),$$
$$\lambda^*(B_L(m)) = m^{3/4}, \qquad \lambda^*(\langle e,h,l\rangle) = (ehl)^{3/4}.$$

The constructions in rows A–D, F, the two scalar-target rows, the single-block row, the T_m rows, CW and zero/rest of Table 1 were proposed and proved by Codex. The scope discipline in the table is also Codex's. We supplied P^* as the witness; it had been found earlier and independently in [2].

We also supplied the unequal-h chain and Kronecker rows; the equal-h chain and its tangent-line argument are already in [2] ("gadget barrier"). We independently re-checked every row as indicated. The transcripts are reproduced in [5].

Theorem 2 (enlarged barrier; Codex and us)**.** *λ^* satisfies every inequality of every family marked **P** in Table 1, for all sizes and parameters. It also satisfies every consequence obtained by adding or multiplying these numerical inequalities, by optimizing the entropy vector q separately for each leg permutation, or by mixing them in a linear program. Hence this system cannot force $t < 3/4$.*

Here "valid" means that the inequality holds for *every* character, via an explicit restriction or degeneration together with additivity, multiplicativity and the shared-leg inequality (paper Cor. 3.2). The validity proofs are in the paper, in [5], or (for odd-k chains) the following weighting. Give the even-position sectors, which are copies of C, weight 0. Give the odd-position sectors, which are copies of C^σ, weight $+1$ on Y and -1 on Z. This generalizes l. 544–548. The "tensor" column records only finite coefficient or support checks, which add evidence but are not proofs. The status column concerns "λ^* satisfies the family". An invalid inequality that λ^* satisfies would not weaken the barrier. The balanced lift turns each nonsymmetrized entropy inequality into its symmetrized form: with all λ_π equal, optimizing $e^{\frac{3}{4}H(q)}\prod_i F_i^{\frac{3}{4}q_i}$ gives $(\sum_i F_i)^{3/4}$.

Table 1: Valid inequality families and the status of "λ^* satisfies them". **P**: proved for all sizes. **E**: exact check on a finite range only. **IC**: interval-certified finite recipe search. Tensor = verification of the underlying degeneration.

id	inequality (source tensor relation)	status	evidence
paper	(C), (T) and their unsymmetrized entropy forms, any q per π	**P**	Thm. 1; balanced-lift reduction
chain	$P(a,\sum_{i\le k}h_i+\frac{(k-1)(a-1)}{2})\ge\sum P(a,h_i)$ (odd k, alternating C/C^σ sectors sharing X; $k=3$, equal h_i is Lemma 4.2)	**P**	tangent line $\phi(a)(b+\frac{a-1}{2})\ge P^*(a,b)$ (equal h_i: [2]); E for $k=3,5$; tensor ($k\le 7$): ours
A	$mP(a,b)\ge\sum_{i\in I}P(a+m-1-2i,b-m+1+2i)$ (via $C(a,b)\otimes B_Z(m)$, apolar filtration)	**P**	Jensen along $a+b$ =const; slack formula re-checked (E); tensor: Codex (780 coeff.)
B	$mP(a,b)\ge\sum_{i<\min(b,m)}P(a,b+m-1-2i)$ (via $C(a,b)\otimes B_X(m)$; $m=2$ is Lemma 4.1)	**P**	Jensen; tensor: ours + Codex
C	$P(a,b)P(c,d)\ge\sum_{i+j=k}P(a+c-1-2i,b+d-1-2j)$ (bivariate D-filtration, shared output)	**P**	proof in Codex round 2; E; tensor: ours + Codex
D	$[P(a,b)P(c,d)]^t\ge\sum_{i<r}P(a+c-1-2i,b+d-1-2i)^t$, $r=\min(a,b,c,d)$ (full direct sum)	**P**	rational certificate $[\,\cdot\,]^3\ge rS^3$; E; tensor: ours
F	$(2M-1)P(Ma,Mb)\ge MP(a,b)$ (Fourier filtering, cost kept)	**P**	monotonicity
O,L	$\max(a,b)\le P(a,b)\le ab$ (outer product; dots)	**P**	$\phi(a)\le 2a/3$
wrap	$P(n,n)^t\ge n$ ($C(n,n)\ge\mathbb{C}[\mathbb{Z}_n]\cong\langle n\rangle$)	**P**	$\phi(n)^3\ge n$
peel	$P(n,n)^t\ge 1+P(n-1,n-1)^t$ ($C(n,n)\rightsquigarrow 1\oplus C(n-1,n-1)$)	**P**	f_n/n non-decreasing; tensor: ours
CW	$P(n,n)^t\ge(27/4)^{1/3}(n-2)^t$ ($C(n,n)\rightsquigarrow CW_{n-2}\ge cw_{n-2}$, standard laser)	**P**	AM–GM; tensor (support): ours
scalar	any tensor product of generators $\rightsquigarrow\langle N\rangle$: $N\le\lambda^*$	**P**	$(d_Xd_Yd_Z)^{1/3}\le P^{*3/4}$
1-block	monomial degeneration (in convolution/group coordinates) of any product of C's, dots and matrix factors to *one* $\langle e,h,l\rangle$	**P**	abelian-group lemma $ehl\le\lvert G\rvert$

$T \to C$	$T_m \rightsquigarrow \bigoplus_s C(a_s, b_s)^{\pi_s}$ (any basis change)	**P**	flattening sandwich, $m^{1/4} > 9/8$
$T \otimes C$	$T_m \otimes C(n,n) \rightsquigarrow \bigoplus C$'s, $m \ge 2$	**P**	sufficient condition (24) of round 2
0/rest	laser on $\{000, 011, 101, 112\}$ partition of $C(n,n)$, any child value $\le$ border rank	**P**	entropy dual; head $q \le 15$ interval, tail rational
Kron	$P(a+(a'-1)N, b+(b'-1)N) \ge P(a,b)P(a',b')$, $N = a+b-1$	E	$a, b, a', b' \le 9$
interval	recursive consecutive-interval laser recipes, $n \le 30$; all root cuts, $n \le 12$	IC	Codex round 3; re-run by us

Remark 3 (open routes). The following are *not* covered by Theorem 2:

- full direct sums of several matrix blocks extracted from powers of $C(n,n)$, especially after non-monomial changes of basis;
- higher-power repartitionings other than the families above;
- $T \otimes C$ relations outside the sufficient condition (for example pure convolution, or very unbalanced (a,b)), and targets that contain products of several C's or matrix factors;
- relations that genuinely use asymmetry (λ_π unequal) at $t = 3/4$.

The uniform quantum functionals give $\mathcal{Q}(C(a,b)) = (ab(a+b-1))^{1/3}$, but on matrix tensors they have total degree 2 rather than 9/4. So they do not close the first route (Codex round 3, §7).

5. Computational certificates

The repository's `scripts/` directory contains four programs, each with its expected output. All decisions use exact rationals, except where a floating-point step is noted.

- `r2_profile_axioms.py` checks (i) exhaustively for $a, b \le 200$ ($b < 800$ for (C)). It also checks the equality cases, the tightness of every step of Lemma 5.1, both induction identities, and Wendel's bound with an interval-certified constant.
- `r2_extension_families.py` is an independent re-implementation of the inequality checks for Table 1, about 0.9×10^6 exact comparisons in total. The 3/4-powers are bracketed by integer fourth roots.
- `r2_tensor_constructions.py` verifies the underlying degenerations at the coefficient level for small sizes. These are the paper's tripling degeneration, the alternating k-sector chains ($k \le 7$), peeling, the CW and zero/rest supports, and the D-adapted bivariate products (7600 basis-pair products) behind B, C, D and Lemma 4.1. A mutation test confirms that the checks detect deliberate errors.
- `r2_least_profile_lp.py` (floating point; not a proof) runs an LP that minimizes $P(n,n)$ under (S), (B), (C), (T) on a $(4n-1)^2$ grid. The minimum agrees with $P^*(n,n)$ to 10^{-13} for $n \le 12$.

References

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